Module 7 Assignment: Concept Generation – Finger Prosthetic

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The brief is a state-of-the-art prosthetic limb for trans-radial amputees, and the first branch point is whether to build it biomimetically — replicating the appearance and the functionality of the hand — or non-biomimetically. Rather than settle that in the abstract, this report decomposes the hand physically, generates both kinds of concept inside every sub-assembly, and lets the answer fall out of the ranking.

Part A — Physical Decomposition and Concept Generation

The system boundary is drawn distal to the socket lamination ring, so the socket, its suspension and the residual limb are interfacing systems rather than parts of the hand [2]. Within that boundary the hand divides into the five sub-assemblies of Figure 1. The split is made so each sub-assembly owns one physical function and can be re-conceived without forcing a redesign of its neighbours [1].

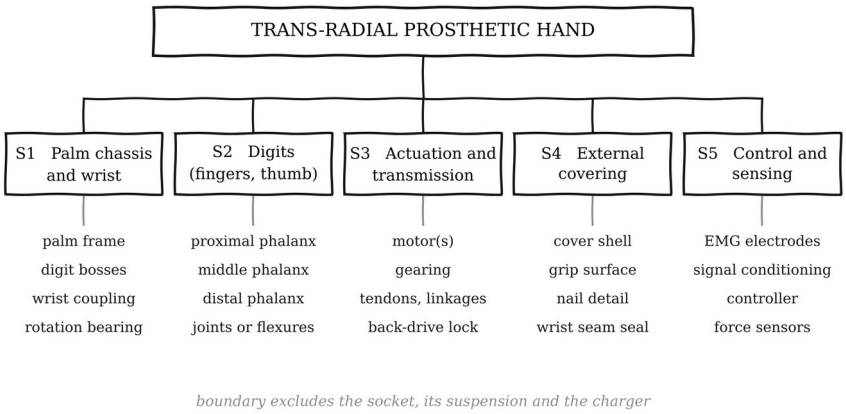


Figure 1. Physical decomposition of the hand into five sub-assemblies.

Each sub-assembly was brainstormed twice. The first pass asked the synectics question — *how does the human hand solve this?* — and produced the biomimetic candidates; the second ran the **SCAMPER** checklist against those results, substituting materials, combining units, eliminating parts and rearranging layouts, which is what produced the monolithic chassis, the whiffletree drive and the exposed shell [1], [8]. Working from two different prompts is deliberate: it guards against the *priming* block, in which an early example quietly fences in everything that follows [1]. The five paragraphs below describe what came out; Figure 2 collects the sketches.

S1, palm chassis and wrist. *A* places the digit bosses on a curved arc with a compliant transverse member, so the palm cups as the digits close, and uses a passive indexed wrist. *B* is a flat chassis machined from one billet with a powered coaxial rotator driving the hand in roll. *C* is a printed lattice carrying bolt-on digit modules on a passive friction wrist.

S2, digits. *A* gives three phalanges in anatomic length ratio with the joints coupled by linkages, so one input curls the digit and the distal joints keep closing after the proximal phalanx stalls on the object, and the thumb sits on a saddle joint. *B* is one moulded part in which notch flexures replace pin joints, reproducing the wrap without the anatomy [7]. *C* is a rigid opposed two-jaw gripper in the split-hook lineage.

S3, actuation and transmission. *A* houses the motors in the forearm and pulls Bowden cables across the wrist in antagonist pairs, as the extrinsic flexors do. *B* puts a sealed gearmotor and non-back-drivable worm set inside each digit, giving five independent axes that hold grip unpowered [4], [5]. *C* drives every digit from one motor through load-balancing bars, so they distribute themselves around whatever shape they meet [7].

S4, external covering. *A* is a custom silicone glove pigment-matched to the sound hand, with nails and fingerprints moulded in. *B* is a moulded elastomer cover chosen for grip friction and ingress protection rather than likeness, replaceable at the wrist seam. *C* has no soft cover at all and makes the anodised structure the exterior.

S5, control and sensing. *A* reads an eight-channel electrode ring and classifies the activation pattern belonging to each intended grasp, so a grip is selected in one step, with fingertip force returned through a vibrotactile band [6]. *B* is the clinical standard: one flexor and one extensor site driving a single open-close axis, with co-contraction stepping through a stored grip list. *C* drops electrodes, selecting grips by wrist gesture or phone app while fingertip sensors regulate slip.

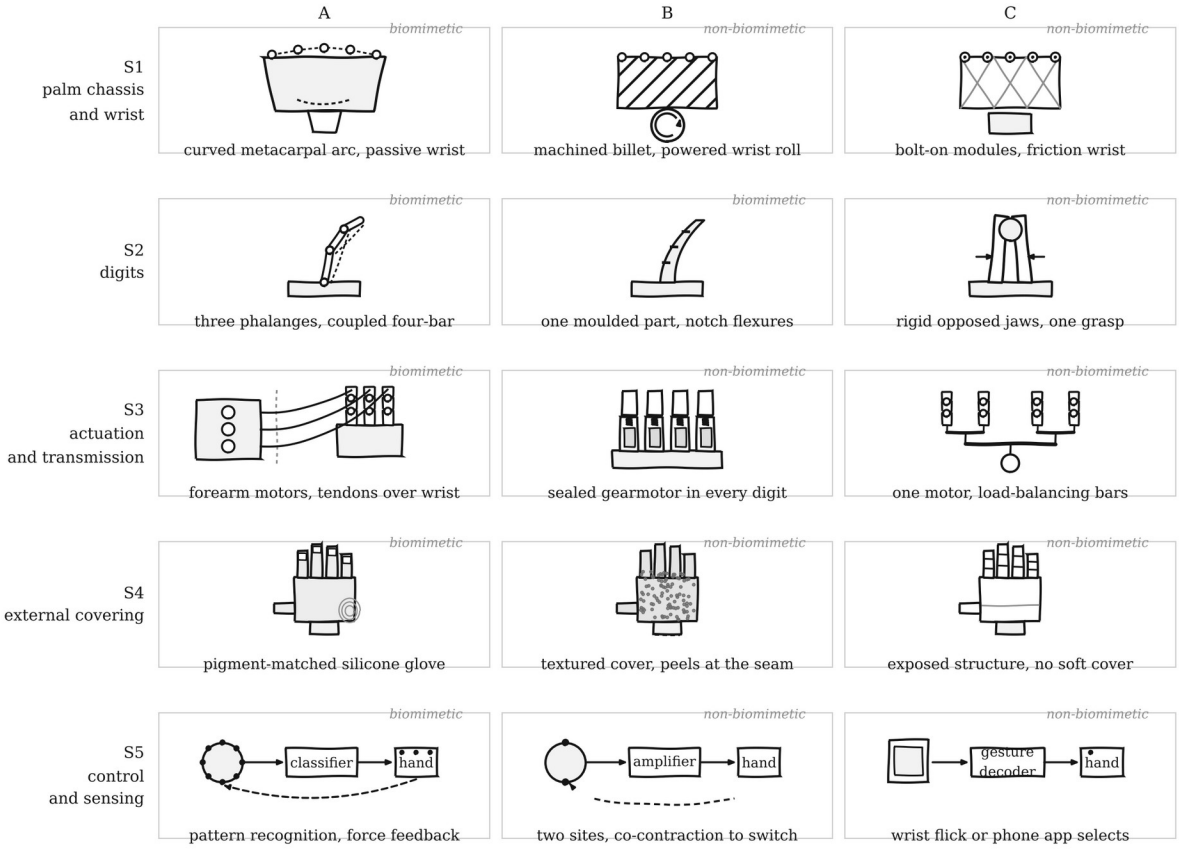


Figure 2. Concept sheet — three concepts per sub-assembly, at least one biomimetic and one non-biomimetic in each row.

Part B — Concept Assessment

Concepts were judged against five weighted criteria, the weights themselves set by a pairwise comparison of the criteria: **functional performance 0.33**, **durability 0.27**, **mass and bulk 0.20**, **cost and complexity 0.13**, **cosmesis 0.07**. Function and weight lead because insufficient function and excess weight are first-order drivers of upper-limb prosthesis rejection, and durability sits above mass because a device in for repair delivers no function at all [3]. Ranking cosmesis last is the most contestable judgement here — appearance genuinely matters to many users [3] — and Part C states what changes if it is wrong.

Concept	Advantages	Disadvantages
S1-A carpal arch	Conforms around irregular objects; natural silhouette; no wrist motor to power.	Compliant member is a fatigue path; wrist set by the sound hand; costliest to tool.
S1-B monolithic	Stiffest and most durable; powered roll restores a lost degree of freedom; simple load path.	Heaviest; adds a motor, driver and current draw; not field-repairable.
S1-C open frame	Lightest and cheapest; digit module swapped in the clinic in minutes.	Least stiff; open structure admits dirt; many fasteners to loosen.
S2-A four-bar	Several distinct grips; high fingertip force; wraps on contact; proportions suit tools built for hands.	Most parts and pin joints; pins wear; tight packaging if a motor must fit inside.

Concept	Advantages	Disadvantages
S2-B flexure	One moulded part, no bearings; inherently safe on impact; lightest and silent.	Stiffness fixed at moulding; elastomer creeps; low pinch force; hard to instrument.
S2-C two-jaw	Very high pinch force per watt; almost nothing to break; cheapest by a wide margin.	One grasp only; cannot wrap or support large objects; least accepted appearance.
S3-A tendons	Low distal mass, so it feels lighter than it weighs; large accessible motors; slim digits.	Cable friction, stretch and fraying dominate servicing; needs forearm volume a short residuum may lack.
S3-B in-digit	Five independent axes; sealed modules; worm holds grip without current; nothing crosses the wrist.	Highest distal mass and inertia; five motors and drivers; a failed motor idles a digit.
S3-C whiffletree	One actuator, fewest parts, lowest cost and power; self-adapts with no sensing.	Digits move together, so grips cannot be selected; no independent thumb; force split set by the linkage.
S4-A silicone glove	By far the best appearance; soft contact; some grip friction.	Elastic resistance loads the actuators continuously; stains and tears; costliest consumable.
S4-B elastomer cover	High grip friction; seals against dust and splash; peels off for replacement; cheap to mould.	Obviously artificial; small mass and actuator penalty; slightly attenuates fingertip force.
S4-C exposed shell	Lightest, cheapest, most serviceable; no actuator load; some users prefer the look.	Mechanism open to dirt and liquid; poor grip on smooth surfaces; pinch points exposed.
S5-A pattern recognition	Selects a grasp in one step; scales to many grips; returns a force channel; hands-free.	Needs training and retraining; sensitive to electrode shift, sweat and limb position; costliest.
S5-B two-site EMG	Simplest, cheapest, best proven; two electrodes to maintain; hands-free.	One axis at a time; every added grip lengthens the co-contraction sequence; tiring and conspicuous.
S5-C gesture and app	No electrodes or skin-impedance problems; menu easy to reconfigure; automatic slip control.	Changing grip needs the sound hand or a phone; fails when both hands are busy; unusable bilaterally.

Table 1. Advantages and disadvantages of all fifteen concepts.

Pairwise ranking	vs A	vs B	vs C	Wins	Rank
S1 A carpal arch	—	0	0	0	3
S1 B monolithic	1	—	1	2	**1**
S1 C open frame	1	0	—	1	2
S2 A four-bar	—	1	1	2	**1**
S2 B flexure	0	—	1	1	2
S2 C two-jaw	0	0	—	0	3
S3 A tendons	—	0	1	1	2
S3 B in-digit	1	—	1	2	**1**
S3 C whiffletree	0	0	—	0	3
S4 A silicone glove	—	0	0	0	3
S4 B elastomer cover	1	—	1	2	**1**
S4 C exposed shell	1	0	—	1	2
S5 A pattern recognition	—	1	1	2	**1**
S5 B two-site EMG	0	—	1	1	2
S5 C gesture and app	0	0	—	0	3

Table 2. Pairwise ranking within each sub-assembly. A 1 means the row concept is preferred to the column concept on the weighted criteria.

What decided each block. In **S1**, powered roll returns a degree of freedom the trans-radial user has genuinely lost, which outweighs the arch's modest conformity gain, and the billet also wins durability. In **S2**, the function gap is at its widest: *C* offers one grasp, and creep in a moulded flexure degrades *B* over the service life. In **S3**, independent axes are what make a multi-grip hand possible, and sealed gearmotors avoid the cable wear that dominates tendon-hand servicing [4]. In **S4**, the glove taxes function continuously and is the costliest consumable, winning only the lightest-weighted criterion. **S5** is the closest call in the study and turns on function alone: two-site EMG cannot address a five-axis hand without co-contraction cycling whose cost grows with every grip added [6].

Ranking in isolation is only half the assessment, because several concepts carry constraints that reach across the decomposition. Two of them are hard incompatibilities rather than preferences.

Coupling	Consequence
S1-B × **S3-A** — Bowden cables cannot cross a continuously rotating wrist without a slip arrangement.	**Hard incompatibility.** Choosing the powered rotator rules out extrinsic tendons; actuation must be intrinsic.

Coupling	Consequence
S2-B × **S3-B** — a monolithic flexure digit has no internal cavity for a gearmotor and worm set.	**Hard incompatibility.** In-digit actuation forces a jointed digit.
S3-C × **S5-A** — a single coupled actuator cannot execute distinct grasps, whatever the classifier outputs.	The control concept has to be matched to the number of independent axes, not chosen separately.
S4-A × **S2-B**, **S3-C** — glove elasticity adds a continuous restoring torque at every joint.	The penalty falls hardest on low-torque underactuated drives, which are exactly the cheap options.
S4-C × **S3-B** — an exposed structure leaves motors, drivers and sensors open to dust and liquid.	The bare shell is admissible only if each digit module is independently sealed.

Table 3. Compatibility couplings between sub-assembly concepts.

The first two rows would have invalidated a naive combination of top-ranked concepts had the rankings fallen differently; as it happens the winning set is self-consistent.

Part C — The Ideal Concept

Combining the first-ranked concept from each sub-assembly gives **S1-B + S2-A + S3-B + S4-B + S5-A**: a monolithic machined chassis with a powered coaxial wrist rotator; five underactuated four-bar digits in anatomic phalanx ratio with the thumb on a saddle joint; a sealed worm gearmotor inside each digit; a replaceable textured elastomer cover; and an eight-channel pattern-recognition EMG interface with fingertip force sensing returned through a vibrotactile band. In use the wearer produces the activation pattern for a grasp and the classifier selects it directly. The digits close under independent control until each stalls against the object, at which point the four-bar coupling keeps curling the distal joints so the hand wraps rather than pinches, and the worm sets hold that grip unpowered.

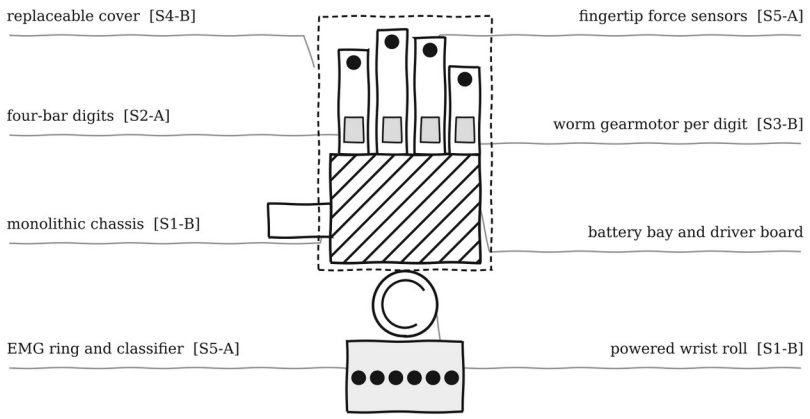


Figure 3. The combined concept, with each element traced to its ranked source.

Figure 4 restates the design in functional rather than physical terms. Each block is phrased so it can be described without an “and” or an “or”, the test that keeps a function pure and lets it be re-allocated to a different physical part later [1]. The mapping is deliberately not one-to-one: F2.2 and F2.3 both land inside one digit module, while F4.1 and F4.2 split between the rotator and the chassis. The functional view is solution-neutral and survives a change of concept; the physical one does not [1].

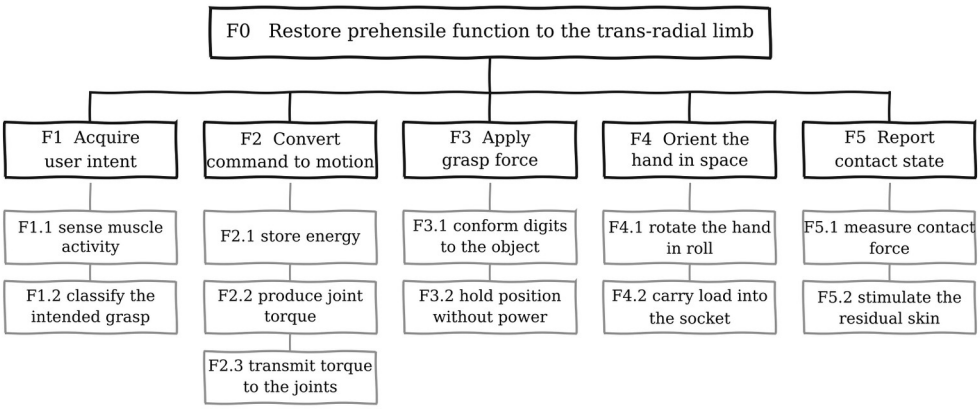


Figure 4. Functional decomposition of the combined concept.

Is the design biomimetic or non-biomimetic?

By count it is non-biomimetic: three of the five winners came from the SCAMPER branch, and one of them, intrinsic in-digit actuation, is the direct opposite of how a hand is built, since the biological hand drives the digits from extrinsic forearm muscles and keeps its mass proximal. The design accepts a real penalty in distal inertia to buy independent digit control. But the count is the less interesting answer. What the ranking actually separated was **biomimicry that buys function from biomimicry that buys only appearance**. The two biomimetic concepts that survived earn their place functionally — anthropomorphic kinematics, because the object world of handles, tools and keyboards was shaped around human hands, and an interface that decodes motor intent and returns tactile information, which mimics the biological control loop rather than the structure. The two ranked out, the carpal arch and the silicone glove, return likeness alone. The honest description is therefore **functionally biomimetic and structurally non-biomimetic**, which is where most current multi-articulating commercial devices have converged [4], [5].

Two limits travel with that verdict. It is **sensitive to the weight vector**: cosmesis was ranked last, and a user who places appearance above mass and cost — a legitimate preference, not an error — would take S4-A and make S1-A competitive, turning the design biomimetic on the count as well. And pairwise ranking is **ordinal**, so it says that B beats C but not by how much; a weighted decision matrix, or a Pugh matrix against the two-site EMG hand as datum, is the natural next step [1].

References

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